

System Requirements Specification

Kei Hea a Nic? Where is Nic?: Dynamic Digital Office Display System

Contents

1 Introduction	5
1.1 Purpose	5
1.2 Project Scope	6
1.3 Overview	7
2 Overall Description	8
2.1 Product Perspective	8
2.1.1 System Interfaces	9
2.1.2 User Interfaces	9
2.1.3 Hardware Interfaces	10
2.1.4 Software Interfaces	11
2.1.5 Communications Interfaces	11
2.1.7 Operations	12
2.1.8 Site Adaptation Requirements	12
2.2 Product Functions	13
2.3 User Characteristics	14
2.4 Operating Environment	15
2.5 Constraints	16
2.6 Assumptions and Dependencies	17
Assumptions	17
Dependencies	17
3 Specific Requirements	18
3.1 Account Creation and Secure Sign In	18
3.1.1 Description and Priority	18
3.1.2 Stimulus/response Sequences	18
3.1.3 Functional Requirements	18
3.2 Availability Viewing and Status Update	19

3.2.1 Description and Priority	19
3.2.2 Stimulus/response Sequences	19
3.2.3 Functional Requirements	19
3.3 Calendar / Timetable View	20
3.3.1 Description and Priority	20
3.3.2 Stimulus/response Sequences	20
3.3.3 Functional Requirements	20
3.4 External Calendar Import	20
3.4.1 Description and Priority	20
3.4.2 Stimulus/response Sequences	20
3.4.3 Functional Requirements	20
3.5 Contact Information Management	21
3.5.1 Description and Priority	21
3.5.2 Stimulus/response Sequences	21
3.5.3 Functional Requirements	21
3.6 Web Application – Diary Server Communication	21
3.6.1 Description and Priority	21
3.6.2 Stimulus/response Sequences	21
3.6.3 Functional Requirements	21
3.7 Data Storage	22
3.7.1 Description and Priority	22
3.7.2 Stimulus/response Sequences	22
3.7.3 Functional Requirements	22
3.8 Serving Data to the Digital Display	22
3.8.1 Description and Priority	22
3.8.2 Stimulus/response Sequences	22
3.8.3 Functional Requirements	22
3.9 Kiosk Mode Operation	23
3.9.1 Description and Priority	23

3.9.2 Stimulus/response Sequences.....	23
3.9.3 Functional Requirements.....	23
3.10 Device Setup and Academic Association.....	23
3.10.1 Description and Priority.....	23
3.10.2 Stimulus/response Sequences.....	23
3.10.3 Functional Requirements.....	23
3.11 Status, Calendar and Contact Views.....	24
3.11.1 Description and Priority.....	24
3.11.2 Stimulus/response Sequences.....	24
3.11.3 Functional Requirements.....	24
3.12 View Navigation	24
3.12.1 Description and Priority.....	24
3.12.2 Stimulus/response Sequences.....	24
3.12.3 Functional Requirements.....	24
3.13 Contact Information Access.....	25
3.13.1 Description and Priority.....	25
3.13.2 Stimulus/response Sequences.....	25
3.13.3 Functional Requirements.....	25
3.14 Status Polling and Automatic Update.....	25
3.14.1 Description and Priority.....	25
3.14.2 Stimulus/response Sequences.....	25
3.14.3 Functional Requirements.....	25
4 Other Nonfunctional Requirements	26
4.1 Performance Requirements.....	26

1 Introduction

1.1 Purpose

The main objective of this System Requirements Specification is to display the requirements for our project “Where is Nic?” Which is a digital dynamic office display system that will be used throughout the School of Computing & Mathematical Sciences at the University of Waikato. This document will be used to ensure that the system will meet function and non-functional requirements whilst also highlighting the interface requirements and capabilities.

The goal of this document is to provide a comprehensive overview of the system for the development team, stakeholders, and client to use as a guide and reference to ensure that the system meets the specified requirements

The intended audiences for this document are:

- **Development Team:** This SRS will serve as a reference for the developers to use to ensure that the system is built according to the requirements and specifications. Should be used to understand the projects scope and overall objectives when developing features for the system as to not deviate from the overarching goals of the system.
- **Users:** Users can use this document to better understand the system to fully comprehend the functionality of the system. Users can also refer to diagrams and specifications in this document to determine if the development team has implemented all the specified requirements.
- **Testers:** The SRS will serve as a guide for testers for the creation of specific test cases that will verify whether the system meets functional and non-functional requirements.
- **Clients:** Clients can use this document to understand the system’s capabilities and limitations whilst also ensuring that it aligns with their needs and objectives for the actual implementation of the system.

This document is structured following the IEEE Standard for Software Requirements Specifications (IEEE 830-1993).

1.2 Project Scope

The dynamic digital office display system digitises the practice of physical paper timetabling and scheduling on the client's office doors. The system is designed to be used by both staff and students to easily access key information regarding timetables and schedules. It modernises an everyday problem through the use of a digital system to improve communication and coordination across the school.

The system requires three integrated components to realise the project scope. Firstly, the system features a web application interface for the staff to access and update the availability information they will display on their office doors. They can customise and update their information in real-time from any internet-connected device. Secondly, the system requires a back-end database that securely manages and stores the availability information between the web application and the display interface. Thirdly, the system features the actual display interface on the client's office doors. This is where users will be able to see the availability information displayed in real-time and be able to interact and scroll through various timetables and schedules to access the information they need.

The system is designed to be user-friendly and intuitive, allowing users to easily and quickly access up-to-date and relevant information. The use of the digital office display system will allow users to access more information than they would be able to with the current physical system whilst being more eye-catching and modern. The system is meant to serve as a full replacement for the current physical system and will be able to be updated and expanded upon in the future with possible integrations with systems like Google Calendar.

1.3 Overview

The SRS is divided into three main sections as follows:

- **Introduction:** This section highlights the scope, purpose and contents of the SRS giving readers an insight into what the document is about and who it's for.
- **Overall Description:** This section displays the overall information and context of the system. It considers the products users and the environment in which the system will be used. It aims to outline the systems functions, constraints, and assumptions and dependencies that will impact the systems requirements. It also includes information regarding the hardware and software interfaces for how the system will operate.
- **Specific Requirements:** This section shapes the system's functionality and performance through stating all the specific requirements. It includes all the necessary information for each individual feature and function whether it is functional or non-functional.

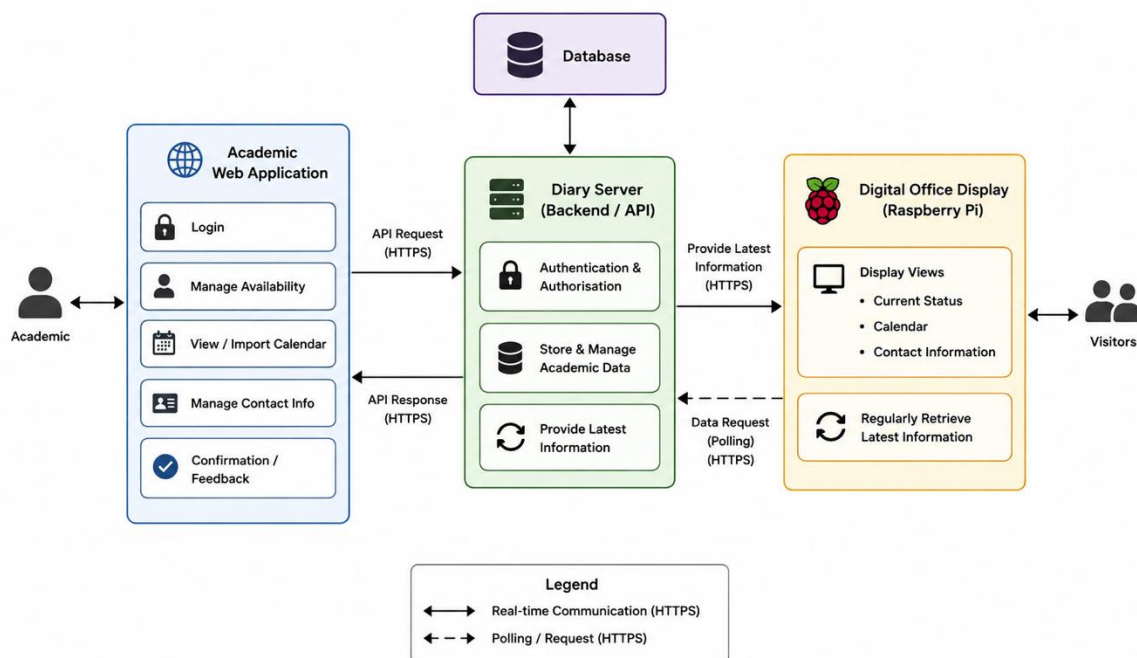
2 Overall Description

2.1 Product Perspective

Our dynamic digital office display system aims to streamline the timetabling and scheduling process that the client currently uses through digitisation. The system will display current availability status, calendar/timetable and contact information for the respective academic.

This will require a web application that is hosted online, that will require login so the staff can update and amend their information that is displayed. This will require 24/7 hosting where we will host using Vercel. This will also then require a back end which will store all the information and authenticate users. We will then also finally need the screen output which will use a Rasbery Pi connected to a small display that will show all the availability information.

This system is completely independent and self-contained using its own database and back-end. This means that for use the users will need to import their own calendar files and data to then be entered into the database as we aren't able to access and use the wider University of Waikato timetabling and scheduling information.



2.1.1 System Interfaces

The system consists of three main components:

- Academic Web Application: Allows academic staff to log in and manage their availability, timetable and contact information.
- Diary Server: Stores academic information and provides the interface between the Web Application and Digital Display.
- Digital Office Display: Retrieves information from the Diary Server and displays the academic's current availability, timetable and contact information to visitors.

The three components communicate through the Diary Server, which acts as the central interface for storing and retrieving system information.

2.1.2 User Interfaces

The system will provide two main user interfaces: the Academic Web Application and the Digital Office Display.

Academic Web Application:

The Academic Web Application will provide academic staff with a web-based interface for managing the information displayed on their Digital Office Display. The interface will include:

- Login Page: Allows academic staff to securely access their account.
- Availability Page: Allows academics to view and update their current availability using predefined statuses such as Available, In a Meeting, Teaching and Out of Office. Academics can also enter an expected return time or custom availability message.
- Calendar Page: Allows academics to view their timetable and import calendar information using an .ics file.
- Contact Information Page: Allows academics to manage the contact information displayed to visitors, including email address, phone number and office location.
- Feedback: The Web Application will provide confirmation when information has been successfully saved and appropriate feedback when an operation fails.

Digital Office Display:

The Digital Office Display will provide a touchscreen interface for visitors to view an academic's information. It will contain three main views:

- **Current Status:** Displays the academic's current availability and any custom availability message.
- **Calendar:** Displays the academic's timetable/calendar information.
- **Contact Information:** Displays relevant contact information and provides a QR code that visitors can scan to access the academic's contact details.

Visitors will be able to navigate between the three views using the touchscreen. The interface will be designed so that the academic's current availability can be understood quickly and from a reasonable distance.

If the Digital Display is unable to retrieve updated information from the Diary Server, it will continue displaying the last successfully retrieved information and indicate that the information may be out of date.

2.1.3 Hardware Interfaces

The Digital Office Display will use a Raspberry Pi connected to a small touchscreen display.

The Raspberry Pi will run the Digital Display application and communicate with the Diary Server through an internet connection. The touchscreen will allow visitors to navigate between the different display views.

2.1.4 Software Interfaces

The system will interface with the following software products:

- Web App Hosting
Name: Vercel
Purpose: Hosts the Academic Web Application and provides access to the application over the internet.
- Database Management System
Name: PostgreSQL
Purpose: Stores academic account information, availability, timetable information and contact information. The Diary Server will communicate with the database to store and retrieve this information.
- Raspberry Pi Operating System
Name: Raspberry Pi OS
Purpose: Provides the operating environment for the Digital Office Display application running on the Raspberry Pi.
- Web Browser to Access Web App
Name: Google Chrome
Purpose: Provides access to the Academic Web Application for academic staff.

The Academic Web Application will communicate with the Diary Server through an API. Requests will contain the required academic information and authentication details and will be transmitted using HTTPS. The Digital Office Display will use the same API to retrieve the information associated with its assigned academic.

Calendar information will be imported using .ics files. The system will process the information contained within these files and store the relevant timetable information in the database.

2.1.5 Communications Interfaces

- Internet Connection: Required for communication between the Academic Web Application, Diary Server and Digital Office Display.
- HTTPS: Used to securely transmit information between the system components.
- API Requests: The Web Application and Digital Display communicate with the Diary Server through authenticated API requests.

2.1.6 Memory Constraints

The Digital Office Display will operate on a Raspberry Pi with limited hardware resources. The application should therefore be designed to operate within the available memory and storage of the Raspberry Pi.

2.1.7 Operations

- Academic Web Application: Academics can log in and update their information when required.
- Diary Server: Continuously stores and provides the latest academic information.
- Digital Office Display: Continuously displays information and periodically retrieves updates from the Diary Server.

If the Diary Server is temporarily unavailable, the Digital Display will continue displaying the last successfully retrieved information until a connection is restored.

2.1.8 Site Adaptation Requirements

Each Digital Office Display must be configured to be associated with a specific academic. This configuration determines which academic's information is retrieved and displayed. This is solved using authentication and login for each display when displaying the staff's information.

The system should allow the Digital Display to be installed in different academic offices without requiring changes to the core application.

2.2 Product Functions

The system provides three main areas of functionality: Academic Web Application, Diary Server, and Digital Office Display.

Academic Web Application:

- **Account Management:** Allows academic staff to securely log in and access their account.
- **Availability Management:** Allows academic staff to view and update their current availability, including their status, expected return time and custom availability message.
- **Calendar Management:** Allows academic staff to view their timetable and import calendar information using .ics files.
- **Contact Information Management:** Allows academic staff to manage the contact information displayed to visitors.

Diary Server:

- **Information Management:** Stores academic account, availability, calendar and contact information.
- **Authentication and Authorisation:** Manages access to academic information and ensures academics can only access their own information.
- **Information Distribution:** Provides the latest academic information to the associated Digital Office Display.

Digital Office Display:

- **Status Display:** Displays the academic's current availability and any custom availability message.
- **Calendar Display:** Displays the academic's timetable and calendar information.
- **Contact Display:** Displays the academic's contact information and provides a QR code for visitors.
- **Information Updating:** Regularly retrieves information from the Diary Server and updates the display when new information is available.
- **Offline Display:** Continues to display the last successfully retrieved information if the Diary Server is temporarily unavailable.

Academic staff manage their information through the Web Application, the Diary Server stores and distributes this information, and the Digital Office Display presents the information to visitors. This is represented by the relationship where Academic Web Application connects to the Diary Server which connects to the Digital Office Display.

2.3 User Characteristics

The System is intended to be used by academic staff and people seeking information about an academic staff member's availability. Users will have different levels of educational background, experiences with university systems, and technical expertise.

- **Academic Staff:** Academic Staff are expected to have basic computer literacy and familiarity with web browsers but are not required to have in-depth and specialist technical knowledge to use the system. This group includes professors, lecturers, teaching fellows, and postgraduate research students. Academic staff are likely to have varying levels of experience with digital scheduling systems and may have limited time available to manage their availability of information. These characteristics provide a basis for considering usability and efficiency.
- **Students:** Students are expected to have varying levels of technical expertise and experience. Most students will have regular experience using web applications through computer and mobile devices, although their familiarity with specific systems may differ. They are likely to use the system to find and access information regarding an academic staff member's availability, meaning they may have little or no prior expertise with the system. Because of this the requirements should take into consideration the manner that the information is displayed.
- **University Staff, Visitors and Guests:** University staff, visitors, and guests may also view an academic staff member's availability. Their familiarity with the university's systems and technical expertise may vary significantly, particularly for visitors who may have no prior experience with the system or are from overseas and are still adjusting. Therefore, the requirements should take into consideration how the information is presented to users who have no prior knowledge of the system.

2.4 Operating Environment

The proposed system consists of a web-based Diary Owner Application, a remotely hosted backend and database, and a Raspberry Pi based Digital display. These components will communicate over a network to allow information to be retrieved and displayed outside their office.

- **Web Application:** The Diary Owner Application will operate as a web-based application accessed through a modern web browser. It will be hosted on a cloud platform and act as a sort of dashboard from where academic staff will be able to manage their information.
- **Cloud backend and Database:** The web application will communicate with a remotely hosted backend that will have an Api that will be used to manage and retrieve system information. A PostgreSQL database hosted using a cloud database service such as Neon will store the information.
- **Raspberry Pi:** The Digital display interface will operate on a Raspberry Pi connected to a monitor positioned outside an academic office. The Raspberry Pi will run the display application in kiosk mode. Additionally, it will run its own Node.js Server which will communicate with the backend API to retrieve the latest availability and timetable information. The retrieved information will then be displayed on the connected display.
- **Network Connection:** The web application, cloud backend, database and Raspberry Pi will rely on network connectivity to manage, retrieve and exchange information.
- **Web Browser:** The Diary Owner Application will be accessed using a modern web browser on a computer or a mobile device. The system is intended to operate independently from a specific operation system if a modern web browser is available.

2.5 Constraints

The following constraints restrict the design and implementation of the proposed system.

- The System must consist of the three integrated components defined within the project scope: the Diary Owner Web Application, backend/API, and Digital Display interface.
- The Digital Display interface must be capable of running on the Raspberry Pi supplied for the project and operating with the monitor installed outside the academic office.
- The Diary Owner Web Application must be web based so that academic staff can access it without requiring the installation of specialist software on their devices.
- The Raspberry Pi will operate as a dedicated display and will have limited hardware resources. The Digital Display interface must be designed with these limitations in mind.
- Communication between the components will be reliant on network connectivity. Therefore, the system must account for the possibility of temporary network interruptions when retrieving updated information.
- The project is intended to produce a prototype rather than a fully production-ready system. The design and implementation will therefore be constrained by the resources, time, and infrastructure available.

2.6 Assumptions and Dependencies

Assumptions

- The Client will provide the Raspberry Pi and monitor required for the Digital display
- The Raspberry Pi will be able to access a suitable network connection.
- The academic staff will have access to a suitable device with a modern web browser to access the Diary Owner Web Application
- The information displayed on the Display interface is suitable for viewing by students, staff, visitors, and other people.

Dependencies

- The Digital display interface depends on the Raspberry Pi and the corresponding monitor being available and operational.
- The Digital display interface depends on a network connection being available to retrieve updated information from the remotely hosted backend.
- The Diary Owner Web Application depends on the availability of the hosting platform on which the application and the backend are deployed.
- The backend depends on the availability of the database service to store and retrieve information.

3 Specific Requirements

This section identifies the functional requirements for the “Kei Hea a Nic? Where is Nic?” system using a feature-by-feature structure: each feature is described, given a priority, illustrated with a stimulus/response sequence, and broken down into individual functional requirements.

3.1 Account Creation and Secure Sign In

3.1.1 Description and Priority

Academic staff must be able to create an account and securely log in to the Web Application so that only the account owner can view and update their own availability, calendar and contact information.

Priority: Must

3.1.2 Stimulus/response Sequences

Stimulus: An academic navigates to the Web Application and enters their registration or sign-in credentials.

Response: The system creates an account, or authenticates the user, and grants access to their own diary data.

3.1.3 Functional Requirements

REQ-1 The system shall allow an academic to create an account and securely log in to the Web Application.

3.2 Availability Viewing and Status Update

3.2.1 Description and Priority

Academic staff must be able to view their current availability and quickly update it using predefined statuses, with the option to specify an expected return time.

Priority: Must

3.2.2 Stimulus/response Sequences

Stimulus: The user opens the availability panel to view their current status, and/or selects a predefined status or enters an expected return time.

Response: The system displays the academic's current availability and updates it based on the user's selection, ready to be shown on the Digital Display.

3.2.3 Functional Requirements

REQ-1 The system shall allow the academic to view their current availability within the Web Application.

REQ-2 The system shall allow the user to update their current availability using predefined statuses (e.g. Available, In a Meeting, Teaching, Out of Office).

REQ-3 The system shall allow the user to enter an expected return time (e.g. "back in 10") to accompany their status.

3.3 Calendar / Timetable View

3.3.1 Description and Priority

Academic staff need to view their timetable/calendar within the Web Application. For the initial prototype (MVP), a single simplified view (e.g. weekly) will be provided rather than separate day/week/month modes.

Priority: Must (calendar view)

3.3.2 Stimulus/response Sequences

Stimulus: The user opens the calendar/timetable section of the Web Application.

Response: The system displays the academic's timetable/calendar using a simplified view suitable for the MVP.

3.3.3 Functional Requirements

REQ-1 The system shall allow the user to view their timetable/calendar within the Web Application.

REQ-2 For the MVP, the system shall present the calendar using a single simplified view (e.g. weekly), rather than separate day/week/month modes.

3.4 External Calendar Import

3.4.1 Description and Priority

Academic staff should be able to bring existing calendar data into the system by importing a standard calendar file.

Priority: Should

3.4.2 Stimulus/response Sequences

Stimulus: The user selects the import option and uploads an .ics file from an external calendar.

Response: The system parses the .ics file and adds the resulting events to the academic's calendar.

3.4.3 Functional Requirements

REQ-1 The system shall allow the user to import calendar information using an .ics file.

3.5 Contact Information Management

3.5.1 Description and Priority

Academic staff must be able to manage the contact information that is shown to visitors on the Digital Display.

Priority: Must

3.5.2 Stimulus/response Sequences

Stimulus: The user updates their contact details (e.g. email, phone number, office location) within the Web Application.

Response: The system saves the updated contact information for display to visitors.

3.5.3 Functional Requirements

REQ-1 The system shall allow the user to manage the contact information shown to visitors (e.g. email, phone number, office location).

3.6 Web Application – Diary Server Communication

3.6.1 Description and Priority

All communication from the Web Application to the Diary Server must be authenticated, and any change the academic makes must be automatically reflected on the Digital Display.

Priority: Must

3.6.2 Stimulus/response Sequences

Stimulus: The user submits a change (e.g. status update, profile edit) from the Web Application.

Response: The Web Application sends an authenticated request to the Diary Server, and the change is automatically reflected on the academic's Digital Display.

3.6.3 Functional Requirements

REQ-1 The system shall communicate with the Diary Server using authenticated API requests.

REQ-2 Changes made through the Web Application shall be automatically reflected on the academic's Digital Display.

3.7 Data Storage

3.7.1 Description and Priority

The Diary Server must receive updates from the Web Application and store all account, availability, calendar and contact information.

Priority: Must

3.7.2 Stimulus/response Sequences

Stimulus: The Web Application sends an update, or an academic's account is created.

Response: The Diary Server receives the update and stores the corresponding account, availability, calendar or contact information.

3.7.3 Functional Requirements

REQ-1 The system shall receive updates from the Academic Web Application.

REQ-2 The system shall store academic account information.

REQ-3 The system shall store academic availability/status information.

REQ-4 The system shall store calendar information.

REQ-5 The system shall store the contact information required by the Digital Display.

3.8 Serving Data to the Digital Display

3.8.1 Description and Priority

The Diary Server must provide the latest academic information to the Digital Display and allow it to retrieve that information on a regular basis.

Priority: Must

3.8.2 Stimulus/response Sequences

Stimulus: The Digital Display sends a request for the latest information.

Response: The server provides the academic's current status, calendar and contact information, and continues to allow the display to retrieve updates regularly.

3.8.3 Functional Requirements

REQ-1 The system shall provide the latest academic information to the Raspberry Pi Digital Display.

REQ-2 The system shall allow the Digital Display to regularly retrieve the latest information.

3.9 Kiosk Mode Operation

3.9.1 Description and Priority

The Digital Display must run as a full-screen kiosk application on the Raspberry Pi.

Priority: Must

3.9.2 Stimulus/response Sequences

Stimulus: The Raspberry Pi is powered on.

Response: The application launches and runs full-screen in kiosk mode.

3.9.3 Functional Requirements

REQ-1 The application shall run as a full-screen kiosk application on the Raspberry Pi preventing tampering.

3.10 Device Setup and Academic Association

3.10.1 Description and Priority

The Digital Display must be associated with a specific academic during initial setup and must remember that association thereafter.

Priority: Must

3.10.2 Stimulus/response Sequences

Stimulus: The device is set up for the first time.

Response: The application associates itself with the specified academic and retains that association so it is remembered on subsequent use.

3.10.3 Functional Requirements

REQ-1 The application shall be associated with a specific academic during initial setup.

REQ-2 The application shall remember which academic it is associated with.

3.11 Status, Calendar and Contact Views

3.11.1 Description and Priority

The Digital Display must show the academic's current availability/status (including any accompanying message), timetable/calendar, and contact information, through three main views.

Priority: Must

3.11.2 Stimulus/response Sequences

Stimulus: The display retrieves the academic's latest information, or a visitor selects a view.

Response: The relevant view: Current Status, Calendar, or Contact Information - is shown with the corresponding up-to-date information.

3.11.3 Functional Requirements

REQ-1 The application shall display the academic's current availability/status, including any accompanying availability message.

REQ-2 The application shall display the academic's timetable/calendar.

REQ-3 The application shall display the academic's relevant contact information.

REQ-4 The application shall provide three main views: Current Status, Calendar, and Contact Information.

3.12 View Navigation

3.12.1 Description and Priority

Visitors must be able to navigate between the Digital Display's views using the touchscreen.

Priority: Must

3.12.2 Stimulus/response Sequences

Stimulus: A visitor interacts with the touchscreen (e.g. by swiping or tapping).

Response: The application transitions to the selected view.

3.12.3 Functional Requirements

REQ-1 The application shall allow visitors to navigate between the Current Status, Calendar and Contact Information views using the touchscreen.

3.13 Contact Information Access

3.13.1 Description and Priority

Visitors viewing the Contact Information view must be able to access the academic's contact details via a QR code or other method.

Priority: Must

3.13.2 Stimulus/response Sequences

Stimulus: A visitor navigates to the Contact Information view.

Response: The application displays a QR code (or other method) that resolves to the academic's contact details.

3.13.3 Functional Requirements

REQ-1 The application shall provide a QR code, or other method, on the Contact Information view so visitors can access the academic's contact details.

3.14 Status Polling and Automatic Update

3.14.1 Description and Priority

The Digital Display must regularly retrieve the latest information from the Diary Server and update automatically when new information is available.

Priority: Must

3.14.2 Stimulus/response Sequences

Stimulus: A scheduled retrieval interval elapses and new information is available on the server.

Response: The application retrieves the latest information from the Diary Server and updates the display automatically, without requiring a manual refresh or restart.

3.14.3 Functional Requirements

REQ-1 The application shall regularly retrieve the latest information from the Diary Server.

REQ-2 The application shall automatically update the display when new information is available.

4 Other Nonfunctional Requirements

4.1 Performance Requirements

Loading Times

- The Academic Web Application pages that being the Login, Availability, Calendar, and Contact Information pages should load and become interactive within 3 seconds of the request under normal network conditions.
- Importing and processing an .ics calendar file should complete within 10 seconds for a typical semester timetable. If processing exceeds 2 seconds, the system should indicate to the academic that the import is in progress.

Response Times

- The Saving of an availability update (status, expected return time, or custom message) should return a success or failure of confirmation within 2 seconds of submission.
- Navigating between the three views on the Digital Office Display (Current Status, Calendar, Contact Information) via the touchscreen should respond within 1 second.

Update Propagation

- The Digital Office Display should retrieve the latest information from the Diary Server at a regular interval of no more than 15 seconds, so that a change made in the Web Application is reflected on the display within approximately 15 seconds.
- When an indication is made that the update has been saved, the user should be able to trust that the update would reach the display within this interval.

4.2 Capacity

- The system will be able to support multiple academic accounts, with each account associated with its own Digital Office Display.
- The prototype, however, will be able to handle 1 academic account without performance degradation.
- The Diary Server will support at least 1 concurrent Web Application session.
- The database should be able to store each academic's availability, contact information, and a full semester of calendar events per account.

4.3 Design Constraints

- The Digital Office Display application must operate within the limited memory, storage, and processing resources of the Raspberry Pi.
- The system will be self-contained, using its own database and back-end. It will not integrate with, or require access to, the wider University of Waikato timetabling and scheduling systems; calendar data is supplied by the academic via .ics import.
- The webserver and networking components must operate within the provisions and constraints of the University of Waikato IT environment.
- The system is delivered as a prototype intended to demonstrate feasibility rather than a production-ready deployment.
- If the Diary Server is unavailable, the Digital Office Display must continue to show the last successfully retrieved information and clearly indicate that it may be out of date.

4.4 Compatibility

- The Academic Web Application should be operable on the latest versions of Google Chrome and should function on other current mainstream browsers.
- The Web Application should be responsive and remain usable across the range of devices academics may use to update their information, including desktop, laptop, tablet, and mobile screen sizes.
- The Digital Office Display application should run on Raspberry Pi Operating System and adapt to the resolution of the connected office touchscreen.

4.5 Interoperability

- The system must accept calendar data in the iCalendar standard format and correctly extract the relevant timetable information.
- The communication between the Web Application, Diary Server, and Digital Office Display must use HTTPS, with data exchanged through the system's API in a standard structured format.
- The QR code presented on the Digital Office Display must encode the academic contact details in a standard, widely readable format so that it can be scanned by common mobile device cameras.

4.6 Security Requirements

User Authentication and Authorization

- Academic staff must log in to the Academic Web Application before viewing or updating their information.
- Passwords must be stored in encrypted form; plaintext passwords must never be stored.
- The Diary Server must ensure that an authenticated academic can only access and modify their own availability, calendar, and contact information.

Data Protection

- All information transmitted between system components must be sent over HTTPS.
- Only the information intended for public viewing (current availability, timetable, and contact details) should be shown on the Digital Office Display; account credentials and management functions must not be accessible from the display.

4.7 Reliability

- The system must minimize data loss. Availability and calendar information successfully saved by an academic must persist in the database.
- The Digital Office Display must degrade gracefully: a temporary loss of connection to the Diary Server must not blank off the screen, and the display must resume showing current information once the connection is restored.
- The Academic Web Application should be hosted so that it is available on a continuous 24/7 basis.
- Backups of the source code should be maintained, preferably on GitHub.

4.8 Usability

- The Digital Office Display must present the academic's current availability so that it can be understood quickly and read from a reasonable distance.
- The display should support accessibility needs, including larger, adjustable text and a high-contrast, color-blind-friendly color scheme.
- The Academic Web Application must provide clear confirmation when information is saved successfully, and informative feedback about the error when an operation fails.
- User interfaces on both the Web Application and the Digital Office Display must allow easy navigation with minimal to no training.
- Text and layout must be clear and readable under normal office lighting and from far away.